

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO-2 : Apply CSS layout and styling techniques to create visually appealing and responsive web interfaces, and use JavaScript to enable interactive client-side behavior. (L3)

Assessment Tool : Short Answer Test

Weightage: 10%

QUESTIONS:

Total Marks: 20

Question 1 (5 Marks)

Suppose that a company wants its website to work seamlessly on mobile, tablet, and desktop devices. Explain the concept of responsive web design and its importance. Describe the key CSS techniques used to achieve responsiveness, and explain how CSS Flexbox or Grid can be used to design adaptive layouts. Illustrate your answer with a simple example layout and suggest improvements to enhance user experience across different devices.

Question 2 (5 Marks)

Suppose that a webpage contains a form where user input must be validated before submission. Explain the concept of client-side validation and its purpose. Describe how JavaScript event handling is used in validation and explain the role of regular expressions in validating inputs such as email or phone numbers. Provide a suitable example and suggest improvements for better usability and security.

Question 3 (5 Marks)

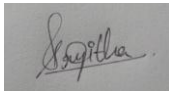
Suppose that a webpage layout breaks when the browser window is resized. Explain the CSS box model and its components, and describe how CSS positioning techniques affect layout structure. Identify possible causes of layout breaking and explain how media queries can be used to resolve these issues. Suggest best practices to ensure a stable and responsive layout.

Question 4 (5 Marks)

Suppose that a website requires dynamic content updates without reloading the entire page. Explain the concept of the Document Object Model (DOM) and its role in web development. Describe how JavaScript can be used to manipulate the DOM for updating webpage content dynamically. Provide an example and suggest improvements to enhance performance and user experience.

Code of Conduct:

I S.Srijitha(192471024) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with	Incorrect or irrelevant answer
				noticeable errors	

Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

CO₂ - Assessment tool-3 [CSA4305]Question 1→ Responsive web Design (RWD)

Responsive web Design (RWD) is a web design approach that makes a website automatically adjust its layout and content according to the screen size of the device. Whether the user opens the website on a mobile phone, tablet, laptop, or desktop, the website should look attractive and easy to use.

For example, an online shopping website should display product in a single column on a mobile app, two columns on a tablet or multiple columns on a desktop.

→ Importance of Responsive web Design.

- * Provides a better user experience on all devices.
- * Eliminates the need for separate mobile and desktop websites.
- * Helps improve SEO (Search Engine Optimization).
- * Reduces maintenance cost because only one website is managed.

→ CSS Techniques used for Responsiveness.1. Flexible layouts

Instead of fixed widths (pixels), use percentages so that elements automatically adjust according to screen size.

Example:

```
.container {  
  width: 100%;  
}
```


2. Flexible Images:-

Images should resize based on the screen size.

Example:-

```
img {  
  max-width: 100%;  
  height: auto;  
}
```

3. media queries:-

media queries allow different CSS styles for different screen size.

Example:-

```
@media (max-width: 768px) {  
  body {  
    font-size: 16px;  
  }  
}
```

→ Using CSS flex box for adaptive layout:-

flexbox helps arrange webpages elements in rows or columns and automatically adjust them based on the available screen space.

Example code:-

```
<div class="container">  
  <div class="box"> home </div>  
  <div class="box"> About </div>  
  <div class="box"> contact </div>
```

```
</div>
```

→ Desktop:-

Header		
menu	content	slide bar
Footer		

→ Mobile:

Header
menu
Content
sidebar
Footer.

→ Improvement for Better user Experience.

- * Provides a better user experience
- * Makes website accessible on all devices.
- * Improves SEO & increases user engagement & satisfaction.
- * Reduces maintainances by using a single website for all devices.
- * Use responsive navigation menu for mobile users.
- * Test the website on different devices & browser image for faster loading.

Question 2:

→ Client - side validation.

The process of checking user input in the browser before the form is submitted to the server. It helps ensure that users enter valid and complete information.

→ Purpose of Client - side validation:

- * Prevents invalid or empty data form being submitted.
- * Provides instant feedback to users
- * Reduces unnecessary server requests.
- * Improves user experience by displaying errors immediately.

→ Java script event handling in validation.

Java script uses events such as `onsubmit`, `oninput`, `onchange` and `onclick` to validate user input. when the user submits the form, java script checks whether the entered data is valid. If the data is incorrect, an error message is displayed and the form submission is stopped.

→ Role of Regular Expression.

- * **Email:** checks whether the email follows the correct format (eg; `abc@gmail.com`).
- * **Phone number:** checks whether the phone number contains exactly 10 digits.

Example:

```
function validateForm() {  
    let email = document.getElementById("email").value  
    let emailPattern = /^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$/;  
  
    if (!emailPattern.test(email)) {  
        alert("Please enter a valid email address.");  
        return false;  
    }  
    return true;  
}
```

→ Improvements for better usability and security.

- * Display clear error messages near the input field.
- * highlight invalid fields. HTML5 validation along with javascript.
- * use HTTPS to protect user data during transmission.

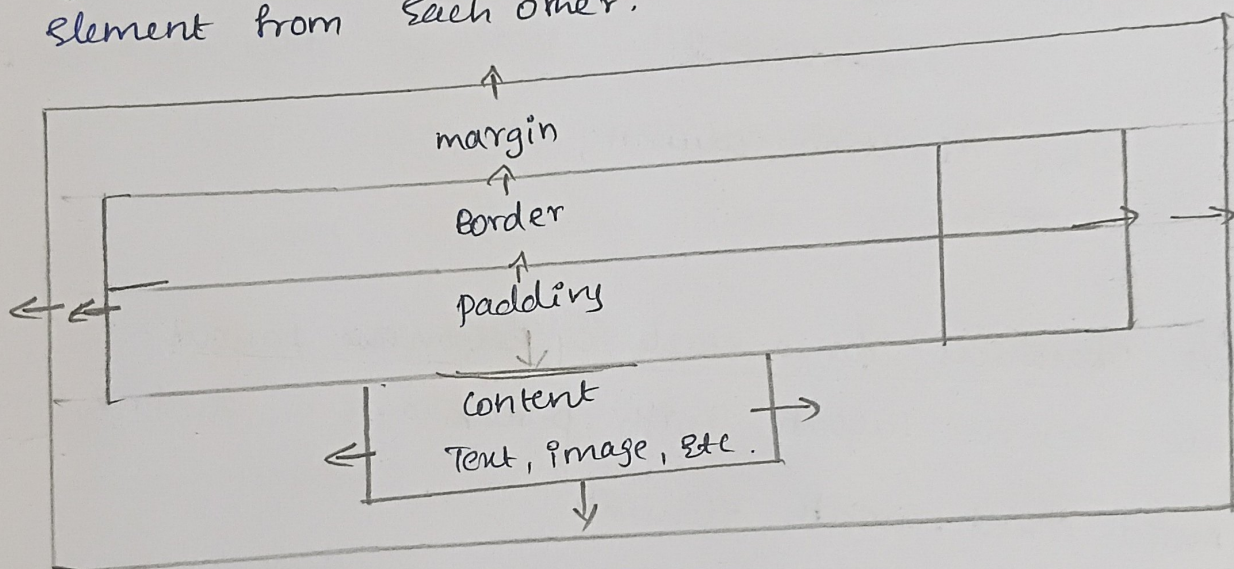
→ Question 3

→ CSS Box Model & Responsive layout.

When a webpage layout breaks after resizing the browser window, it is usually due to fixed sizes, incorrect positioning or missing responsive design techniques.

→ CSS Box Model.

1. Content - The actual text or image inside the element.
2. Padding - The space between the content and the border.
3. Border - The line that surrounds the padding and content.
4. Margin - The space outside the border that separates element from each other.



→ CSS positioning Techniques.

Static: Default position

Relative: Move an element relative to its normal position.

Absolute: Positions an element relative to its nearest position parent

Fixed: Keep the element fixed on the screen while scrolling.

Sticky: Acts like relative until scrolling reaches a certain points.

→ Causes of Layout Breaking.

- * fixed width elements (using pixels instead of Percentages)
- * large images that do not resize
- * incorrect use of CSS positioning
- * missing media queries
- * overflowing content or long text.

→ using media queries.

Example:

```
@media (max-width: 768px) {  
  .container {  
    flex-direction: column;  
  }  
}
```

→ Best Practices for a stable & Responsive Layout.

- * use flexible layouts with percentages or flexbox/grid.
- * Avoid fixed-width elements
- * make images responsive using max-width: 100%.
- * use media queries for different screensize.
- * Test the website on mobile, tablet, and desktop devices
- * maintain proper spacing, margins and padding.

Question - 4

→ Document Object Model (DOM)

The Document Object Model (DOM) is a programming interface that represents an HTML webpage as a tree of objects. It allows JavaScript to access, modify, add or remove HTML elements without reloading the entire webpage.

→ Role of DOM in web Development.

- * Represents HTML elements as objects.
- * Allows JavaScript to access and modify webpage content.
- * Enables dynamic updates without refreshing the page.
- * Improve user interaction and responsiveness.
- * Supports creating interactive web application.

→ JavaScript DOM Manipulation.

- getElementById()
- querySelector()
- innerHTML
- textContent.

Example:

HTML:

```
<P id = "demo" > welcome! </P>  
<button on click = "changeText()" > click
```

JavaScript:

```
function changeText() {  
    document.getElementById("demo").  
}
```


→ Output:-

When the user clicks the click me button the text changes from "welcome" to "welcome to Javascript DOM!" without reloading the webpage.

→ Improvements for Better performance and user Experience.

- * update only the required elements instead of reloading the entire page.
- * minimize unnecessary DOM manipulations.
- * use event handling efficiently
- * Display loading indicators for long operations.
- * optimize JavaScript code for faster performance.
- * Test the webpage on different browsers and devices.